

# **State Trends in Firearm Ownership and Crimes in the United States from 1980 Through 1999**

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## Introduction

### The Data

We used two datasets in our project: an embedded dataset from r called “MoreGuns, Less Crime?” (Stock & Watson) and a dataset from researchers at RAND’s Gun Policy Initiative called State-Level Firearm Ownership (Schell et al., 2020). The datasets were merged in order to provide more information about relationships between measures of gun ownership, demographics, and crime rates. Each column is a predictor. Each row is a set of observations within a specified year and state. To avoid the 49 dummy variables that would be produced by a single-state predictor variable, we used the US Census Bureau’s regional map to divide the states into broader geographical/cultural regions (U.S. Census Bureau). Table 1A with variable names and descriptions is included in the appendix.

### A note about the data:

Household Firearm Ratio (HFR) is an estimate created using modeling software in the study performed by RAND researchers. They used other survey data as well as predictors of HFR (which is included in their dataset) to model HFR for all states and years included in the dataset. We have determined it to be an acceptable estimate, but because it was created using other variables in the dataset, there is high correlation between several of the variables and HFR. Specifically, the researchers found that Male Firearm Suicide was an especially strong predictor of HFR, which has shown up in many of our models as collinearity between these two predictors (Schell et al., 2020).

### The Project

There are many misconceptions rooted in ideological notions about gun legislation and the potential relatedness between gun ownership, gun laws, and crime in the United States. An unbiased approach to determining the true relationships between these social factors is needed to dispel these misconceptions and help resolve tensions over extreme political polarization in the United States. We engaged in exploratory analysis to discover any apparent relationships between predictors and gun ownership, gun laws, and crime. We were particularly interested in the numerical response variables of Household Firearm Rate (HFR), violent crime rate, murder rate, and robbery rate. We were also interested in the categorical response variables of the three gun regulation laws in the dataset including universal background checks, shall-carry laws, and gun permit requirements. We fit models to each of these responses and paid special attention to their influence on the other responses when any of these response variables were used as predictors in other analyses. In each model, we were especially interested in the influence of gun ownership, gun control, and proxy predictors related to gun ownership on the specified outcome. Though inference is our main goal, we would also be curious to learn more about which types of models are accurate at predicting our responses.

## Classification Models

The goal of our classification models was to conduct exploratory analysis and make inferences about the relationships between the set of predictors and the presence or absence of the three gun related laws: universal background check laws, shall-carry laws, and permit to purchase laws. A secondary goal was to discover which statistical classification method produced the most accurate predictive models.

Universal background check laws are policies that require all sales of firearms to be conducted through a licensed dealer who must perform a background check on buyers before any weapons are sold. Shall-carry laws, also known as shall-issue laws, require that a state or governing body must not deny any applicant a concealed carry permit as long as the applicant meets all permit criteria. Permit-to-purchase laws require an applicant to receive a permit before they can purchase any firearms.

For each law, the proportion of observations where the law was present was calculated for the whole guns data set, the training data, and the testing data. It was apparent that each of these sets of data were biased heavily towards observations where each law is not present. These proportions can be found in Figure 1.

| Law                     | Prop. of All Data | Prop. of Training Data | Prop. of Testing Data |
|-------------------------|-------------------|------------------------|-----------------------|
| Universal Bkgnd. Checks | 0.151             | 0.1475                 | 0.165                 |
| Shall-Carry             | 0.237             | 0.27375                | 0                     |
| Permit-to-Purchase      | 0.195             | 0.19                   | 0.215                 |

*Figure 1: Proportions of observations where a given law is present for whole data set, training data, and testing data.*

For each law, a value of 0 indicated that the law was not present for the given state and year, while a value of 1 indicated that the law was present. Five classification models – logistic regression, lasso, ridge, Naive Bayes, and KNN– were fit for each law, with the presence or absence of the law serving as the response variable in each model. The logistic regression models were used for interpretation and the remaining models were used in comparison with logistic regression to determine the most accurate predictive model. We will report and interpret the slopes for the three most influential significant predictors in each logistic regression model.

Because our data is biased towards observations that do not have each gun law in place, we will use AUC as a measure to compare the accuracy of the models.

The data was partitioned for training and testing purposes by randomly selecting eighty percent of observations as training data and the remaining twenty percent set aside for testing purposes. This was done to ensure that data points not used to train the models were available to test the predictive capacity of each model.

To select variables to include in the logistic regression models, backwards stepwise selection was performed. After calculating the VIFs for the predictors in these models, further predictors with high VIFs were removed based on their p-values produced in the original backwards selection model. We most often removed variables that had a high VIF and a larger p-value in comparison to the other variables.

We resolved to include all predictors in the Naive Bayes and KNN models as using predictors selected in backwards selection of the logistic models might not be as optimal given the differing assumptions of KNN and Naive Bayes.

### ***Universal Background Check Models***

The three significant predictors with the highest effect in the logistic regression model used to predict the presence or absence of universal background check models were the *HuntLic*, *EastSouthCentral*, and *permit* variables, with slopes 7.86, 5.93, and 5.68 respectively. This model produced a test AUC of 0.9520958 and a test proportion of incorrect classifications of 0.12. To reduce variance inflation, the variables *cauc*, *Male\_FS\_S*, and *robbery* were removed. In order to keep interesting variables in the model, not all VIF values were reduced to below 10. However, this should not affect inference as almost all predictors are already significant; decreasing VIF would only make these predictors more significant (since decreases in the VIF are brought about by lowered variance, lowered variance lowers standard error of the slope, lower standard error of the slope, as the denominator of the expression for T-statistic, grows the T-statistic, a larger T-statistic is associated with a smaller area under the curve of the T-distribution, therefore P-Values for the Null that the slope is 0 are lowered further when the VIF is lowered). To see a full table of predictors associated with a greater likelihood of universal background check laws, see the appendix.

The slopes can be interpreted as follows:

1. Holding all other predictors fixed, the log odds of having a universal background check law present in a state increases by 5.67 when the state has a permit to purchase law in place compared to when it does not. This relationship can be seen below in figure 2.

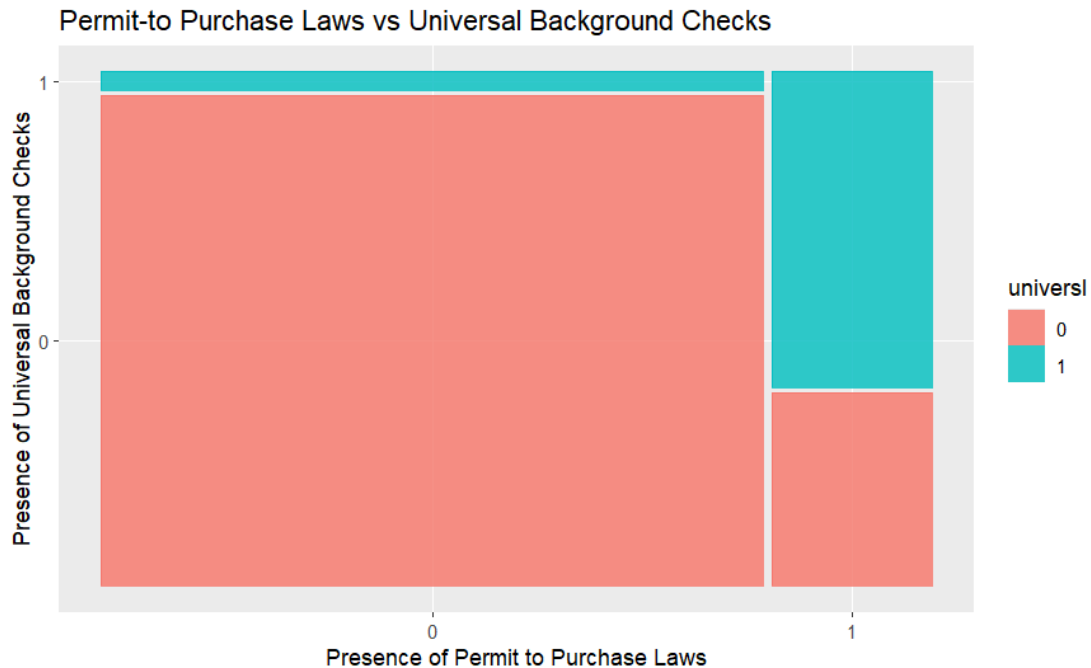


Figure 2: Mosaic plot visualizing relationship between permit to purchase laws and universal background check laws

- Holding all other predictors fixed, the log odds of having a universal background check law increases by 7.86 for each increase in one standard deviation in *HuntLic*.
- Holding all other predictors fixed, the log odds of having a universal background check law present in a state increases by 5.93 when the state is in the east south central region (Alabama, Kentucky, Mississippi, and Tennessee) as compared to when the state is not. This could be because Tennessee had this law present for all years in the data set except for the last year, 1999, and all other states in this region did not have this law present for any of the years. R might be picking up on this difference as significant. However, this law was repealed in Tennessee in the last year, so this interpretation may no longer be accurate for years post 1999.

Additionally, the model that produced the highest AUC was the lasso regression model, with an AUC value of 0.9598984. The percentage of incorrect classifications for this model was 9.5%. See figure 1 for a table of AUC values and proportion of incorrect classifications for the other four models.

| Model | AUC Value | Prop. of Incorrect Classifications |
|-------|-----------|------------------------------------|
|-------|-----------|------------------------------------|

|                        |           |       |
|------------------------|-----------|-------|
| Lasso                  | 0.9598984 | 0.095 |
| Ridge                  | 0.9286881 | 0.135 |
| Naïve Bayes            | 0.9185266 | 0.13  |
| Logistic<br>Regression | 0.9520958 | 0.12  |
| KNN                    | 0.9020142 | 0.02  |

Figure 3: AUC values and proportions of incorrect classifications for universal background check models

### ***Shall-Carry Law Models***

The three significant predictors with the highest relative effect on presence of shall-carry laws in the logistic regression model were *NewEngland*, *Year*, and *permit/EastSouthCentral*, with slopes 5.67, 4.01, and -3.32 respectively. This model produced a test AUC of 0.9134957 and a test proportion of incorrect classifications of 0.16. We removed the *cauc* and *SouthAtlantic* variables to reduce variance inflation.

The slopes are interpreted as follows:

1. Holding all other predictors as fixed, the log odds of having a shall-carry law in place increases by 5.67 for states in the New England region (Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, and Vermont) as compared to states not in the New England region. This is opposite of what might have been expected, as shall-carry laws are generally recognized as pro-gun, a trait uncommon to this region.
2. Holding all other predictors as fixed, the log odds of having a shall-carry law in place increases by 4.01 for each standard deviation increase in year. This might suggest that shall-carry laws simply became more prevalent in more recent years. See figure 4.

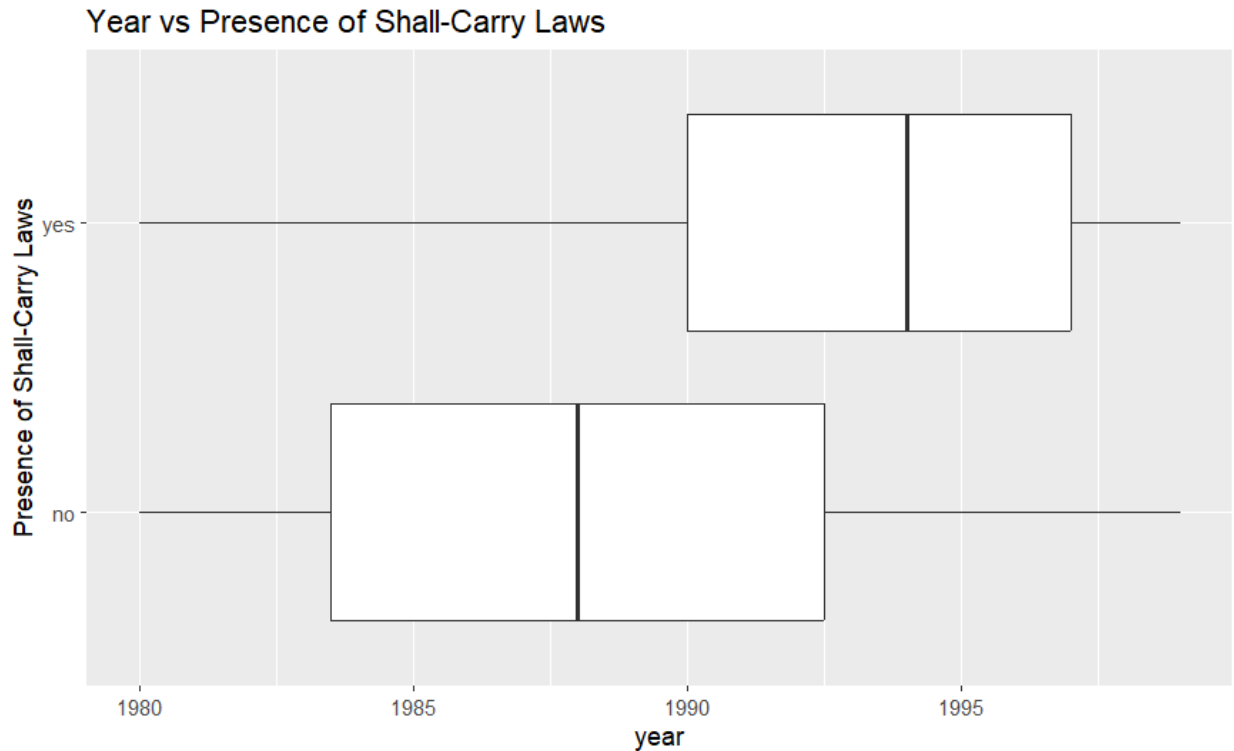


Figure 4: Visualization of the relationship between year and shall-carry laws

3. Holding all other predictors as fixed, the log odds of having a shall-carry law in place decreases by 3.32 when a state has a permit to purchase law in place as compared to states that do not have permit to purchase laws. These results might make sense, as permit to purchase laws are intended to regulate gun usage, whereas shall-carry laws in a sense do the opposite.
4. Similarly, the log odds of having a shall-carry law in place decreases by 3.32 when a state is from the East South Central Region (Alabama, Kentucky, Mississippi, and Tennessee) as compared to otherwise.

Furthermore, the model that produced the highest AUC was the Naive Bayes model, with a test AUC of 0.9288432 and a test proportion of incorrect classifications of 0.165. It is worth noting that the KNN model had a proportion of incorrect classifications that was extremely lower than all other models, at 0.02; however it did have the lowest AUC out of the models. Additionally, the Lasso, Ridge, and logistic regression models all produced the same test proportion of incorrect classifications of 0.16, marginally more accurate than the Naive Bayes model. See figure 5 for a table of all AUC and classification errors for models.

| Model               | AUC Value | Prop. of Incorrect Classifications |
|---------------------|-----------|------------------------------------|
| Lasso               | 0.9147641 | 0.16                               |
| Ridge               | 0.910071  | 0.16                               |
| Naïve Bayes         | 0.9288432 | 0.165                              |
| Logistic Regression | 0.9134957 | 0.16                               |
| KNN                 | 0.867136  | 0.02                               |

Figure 5: AUC values and proportions of incorrect classifications for shall-carry laws

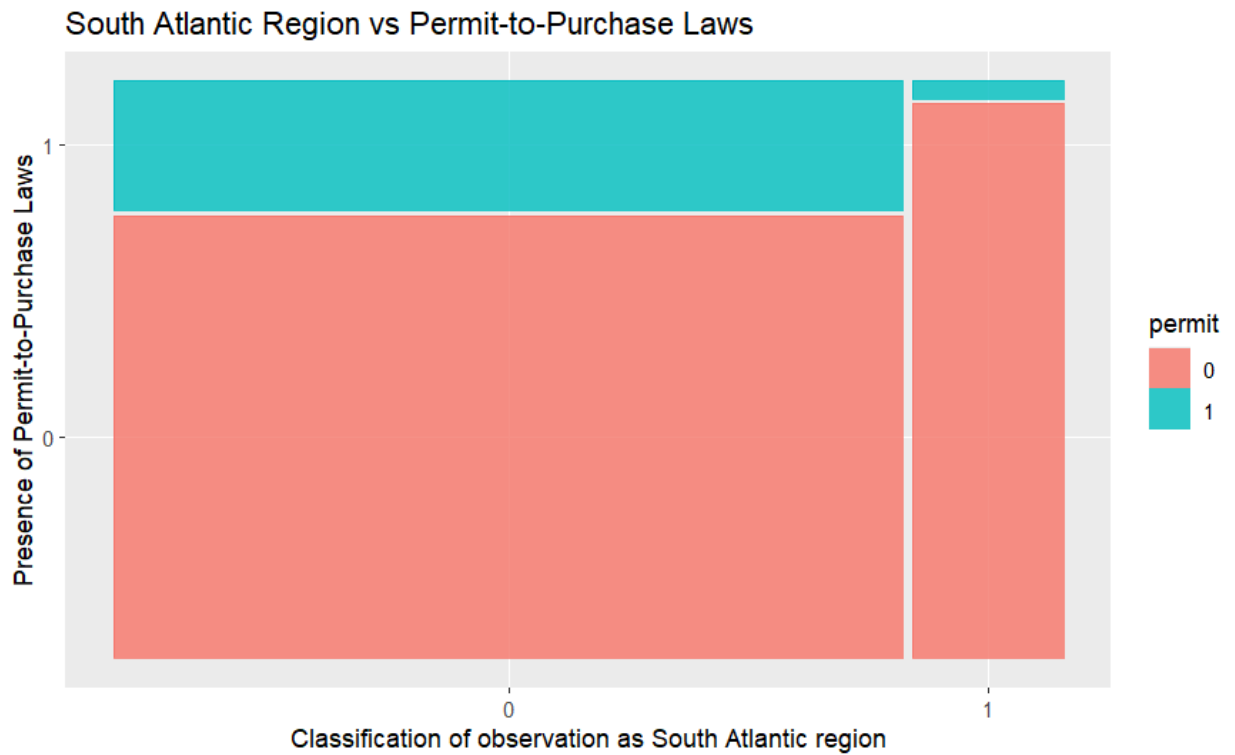
### ***Permit to Purchase Law Models***

The three significant predictors with the greatest influence on the presence of permit to purchase laws for the logistic regression model were *Household Firearm Rate (HFR)*, *Pacific*, and *SouthAtlantic*, with slopes -4.14, -6.06, and -5.59 respectively. This model produced an AUC of 0.9927418 and a proportion of incorrect classifications of 0.02. As a note, the variables *cauc* and *robbery* were removed to lower VIF values for predictors in this model.

The slopes are interpreted as follows:

1. Holding all other predictors as fixed, the log odds of having a permit to purchase law in place decreases by 4.14 with an increase in one standard deviation of the household firearms rate.
2. Holding all other predictors as fixed, the log odds of having a permit to purchase law in place decreases by 6.06 for states that are in the pacific region (Alaska, California, Hawaii, Oregon, and Washington) as compared to states that are not. Considering the cultural and political climates regarding gun laws of states like Alaska in comparison to California, we might question if there is a specific state in this region that has a greater effect on this negative slope.
3. Holding all other predictors as fixed, the log odds of having a permit to purchase law in place decreases by 5.59 for states that are in the south atlantic region (Delaware, Florida, Georgia, Maryland, North Carolina, South Carolina, Virginia, and West Virginia) as compared to states that are not. See figure 6.





*Figure 6: A visual representation of the relationship between the South atlantic region and permit to purchase laws*

The model that produced the highest AUC value was the lasso model with a value of 0.9949637 and an accompanying proportion of incorrect classifications of 0.02. It is worth noting that again, the KNN model produced the lowest proportion of incorrect classifications at 0.005, but the lowest AUC value of 0.9785958. See figure 7 for a complete table of AUC values and proportions of incorrect classifications.

| Model               | AUC Value | Prop. of Incorrect Classifications |
|---------------------|-----------|------------------------------------|
| Lasso               | 0.9949637 | 0.02                               |
| Ridge               | 0.9897793 | 0.02                               |
| Naïve Bayes         | 0.9826692 | 0.07                               |
| Logistic Regression | 0.9927418 | 0.02                               |

|     |           |       |
|-----|-----------|-------|
| KNN | 0.9785958 | 0.005 |
|-----|-----------|-------|

*Figure 7: AUC values and proportions of incorrect classifications for permit to purchase laws*

The logistic regression models for each law all had as one of the three most influential significant variables a region. In terms of statistical inference and interpretation, this might indicate that the presence or absence of a gun law in a state is heavily associated with where the state is regionally in the United States. Possibly conjectures might include that different regions of the US vary culturally and politically, and this variety might lead to adaptations or rejections of different gun policies. Further research could include finding statistical measures of culture or politics for each region and running hypothesis tests with these variables as predictors of law presence. Further action could also include investigating potential differences between states in each region. For example, data showed that whether or not a state was in the pacific region was influential on whether or not the state had a permit to purchase law. We could see if there were certain states, like Alaska, that might have had a greater influence on this result.

It is also visible that the presence of one gun regulation law can be important in predicting the presence of others. For example, we saw that states with permit-to-purchase laws were more likely to have universal background checks. Future research could be done to see if there is a causal relationship between these two variables.

We can also observe that the proportion of incorrect classifications for each model for the permit to purchase law were greatly lower than models for universal background checks and shall-carry laws. This might indicate that our set of predictors is better able to capture the relationship with the presence of permit to purchase laws as compared to the other two, and that the other two laws are harder to predict the presence of, or have a more complicated relationship with the predictors that our models were not able to capture. Additionally, for each of the three laws, the KNN models were able to produce the lowest proportion of incorrect classifications. This could indicate that the decision boundary between present or absent is highly complicated for each law, as all three KNN models performed best with small K values.

Further action could also include conducting feature selection to tune parameters for the KNN and Naive Bayes models. Furthermore, principal component analysis could also be used for the logistic regression models to combine highly collinear variables instead of removing them from the models and further lower VIF values for predictors.

For two out of the three laws, the lasso model produced the lowest AUC value, indicating that this model might be the most suitable if our goal is accurate prediction. However, given we only modeled the presence or absence of three laws, this might be a coincidence and we would need to model more gun laws to confirm if this trend of LASSO being the best predictive model to use with our dataset of legislative predictors holds true for other laws as well.

## Regression Models

We were also interested in predicting four numerical responses: murder rate, violent crime rate, robbery rate, and household Firearm Ratio (HFR). In our regression methods, six different types of models were fitted to the data: linear regression (gaussian identity), gaussian with a log link, gaussian with a square-root link, Gamma with a log link, Gamma with a square-root link, and KNN (used for comparing predictive accuracy, not for interpretation). Models were compared using test MSE. We did not use the poisson GLM family because none of the response rates are whole number counts in the dataset.

With each model, we performed backwards selection to remove variables. LASSO selection was run for each response variable's predictive model and for each type of GLM. In the cases of violent crime and murder response models, forward selection was also attempted. When applied, forward selection arrived at more final variables than backwards selection, making the model harder to interpret, so this method was not applied to the robbery rate or HFR analysis. LASSO regression was producing extremely high lambdas in some instances that made it difficult to turn the results into an interpretable model, so LASSO models have been excluded from this inferential report. After selecting the variables, each model's diagnostic plots and VIFs were tested to ensure that it was possible to interpret the results of the models. With most models, there was a high leverage point that was removed. High VIFs frequently occurred, so the multi-covariate related predictor corresponding with the highest p-value was taken out until the issue was resolved with all VIFs of predictors beneath the threshold of 10. When both variables were present, either "cauc" or "afam" was removed as they both had large VIFs (Values ranging from 50 to 100). Occasionally HFR was causing 3+ high VIFs and was removed.

For murder rate and violent crime rate, the KNN model's variables were selected by choosing the variables that appeared in 4/5 or 5/5 of the GLM models. The KNN models had the smallest MSE by far in predicting both murder and violent crime. Since we are interested in interpretation, this was not as helpful to us, but did show us that our best models are not as predictive as the more flexible KNN that adapts to high levels of variance. We've hypothesized that the KNN performed the best because of its flexibility in high variance situations. Many of the variables included in the models had high Variance Inflation Factors due to relationships between these predictors which became an issue with the GLMs, but was not an issue with the flexible KNNs.

### ***Murder Rate Model***

Regression models were fit to predict murder rate. This is a predictor measured in incidents per 100,000 people, counted by year and state. The murder rate data has a distribution of:

| Min. | 1st Qu. | Median | Mean  | 3rd Qu. | Max. |
|------|---------|--------|-------|---------|------|
| 0.2  | 3.7     | 6.4    | 7.665 | 9.8     | 80.6 |

*Figure 8: Descriptive statistics for the Murder Rate*

To predict murder rate, five GLMs and one KNN were fitted. The test MSEs were as follows:

| Gaussian identity link | Gaussian log link | Gaussian sqrt link | Gamma log link | Gamma sqrt link |
|------------------------|-------------------|--------------------|----------------|-----------------|
| 2.079641               | 2.259565          | 2.360212           | 2.726408       | 2.882427        |

KNN test MSE: 1.38660849331006 at K=3

*Figure 9: Test MSEs for the Murder Rate Regression Models*

The strongest performing interpretable model was the linear, or gaussian identity, model with variables selected using backwards selection. The low MSE of the KNN model indicates that none of these models were as flexible to the high variance of the data, and therefore as high performing, as the KNN. In the linear regression model, the predictor “cauc” was removed due to collinearity with “afam”; “cauc” had the lower p-value. After backwards selection, fourteen predictors were left in the model and all were statistically significant at  $\alpha = 0.05$ . See the table below. The full summary of this model is included in the appendix as Figure 2A.

| Predictors positively associated with murder rate                                                                                                                                                       | Predictors negatively associated with murder rate                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Violent crime rate<br>Robbery rate<br>African American population<br>Male population<br>Population density<br>Male firearm suicide<br>Presence of universal background checks law<br>West South Central | Prisoner population<br>Pacific<br>Mountain<br>West North Central<br>South Atlantic<br>Middle Atlantic |

*Figure 10: Associations of predictors with murder rate*

The intercept of the model is 7.8179928, meaning that when all predictors in the model are 0, the expected murder rate is 7.8179928 per 100,000 people. Because the predictors are standardized, this value is expected when all of the numerical predictors are at their mean and the state is in the East North Central, East South Central, or Northeast region of the United States with no Universal Background Checks law. Slope interpretations for the five strongest predictors follow. A one SD increase in violent crime rate is associated with an increase of 1.7705967 murders per 100,000 people, holding all other predictors constant. A one SD increase in robbery rate is associated with an increase of 2.3448012 murders per 100,000 people, holding all other predictors constant. A one SD increase in male firearm suicide rate is associated with an increase of 1.8681334 murders per 100,000 people, holding all other predictors constant. States in the Pacific are associated with a decrease of 1.3232955 murders per 100,000 people compared to states outside of the Pacific, holding all other predictors constant. States in the South Atlantic are

associated with a decrease of 1.3680404 murders per 100,000 people, holding all other predictors constant.

Notable observations from this data include that the presence of a universal background check law is associated with an increase in murder rates. This may be explained by the higher population density in states that tend to pass gun control laws. A boxplot of this relationship is included in the appendix as Figure 3A, which shows the increase in murder rates in states with a universal background checks law. Further investigations should analyze if the universal background check law is more common in states with certain characteristics (ie. population, political party, demographics) before drawing conclusions about its effectiveness.

### ***Violent Crime Rate Model***

Regression models were fit to predict violent crime rate. This is a predictor measured in incidents per 100,000 people, counted by year and state. The violent crime rate data has a distribution of:

| Min. | 1st Qu. | Median | Mean  | 3rd Qu. | Max.   |
|------|---------|--------|-------|---------|--------|
| 47   | 283.1   | 443    | 503.1 | 650.9   | 2921.8 |

*Figure 11: Descriptive Statistics of Violent Crime*

To predict violent crime rate, five GLMs and one KNN were fitted. The test MSEs were as follows:

| Gaussian identity link | Gaussian log link | Gaussian sqrt link | Gamma log link | Gamma sqrt link |
|------------------------|-------------------|--------------------|----------------|-----------------|
| 7490.159               | 10966.64          | 8628.858           | 13435.4        | 9496.849        |

KNN test MSE: 2443.44867801047 at K=2

*Figure 12: Test MSEs for the Murder Rate Regression Models*

The strongest performing interpretable model was the linear, or gaussian identity, model with variables selected using backwards selection. The predictor “cauc” was again removed due to collinearity with “afam”; “cauc” had the lower p-value. The predictor Male\_FS\_S was also removed due to collinearity with HFR. This removal did not decrease the VIF of HFR to a value below 10 (12.386264) but HFR is a variable of interest and thus was left in the model. After backwards selection, twenty predictors were left in the model and all were statistically significant at  $\alpha = 0.05$ . See the table below. The full summary of this model is included in the appendix as Table 3.

|                                                          |                                                          |
|----------------------------------------------------------|----------------------------------------------------------|
| Predictors positively associated with violent crime rate | Predictors negatively associated with violent crime rate |
|----------------------------------------------------------|----------------------------------------------------------|

|                                    |                                             |
|------------------------------------|---------------------------------------------|
| Year                               | Afam                                        |
| Murder rate                        | Income                                      |
| Robbery rate                       | Presence of universal background checks law |
| Prisoner population                | Female Firearm Suicide                      |
| Population density                 | Hunting Licenses                            |
| Household Firearm Ratio            | <i>Guns &amp; Ammo</i> Subscriptions        |
| Presence of permit to purchase law | East South Central                          |
| Pacific                            | Middle Atlantic                             |
| Mountain                           | Presence of shall carry law                 |
| West North Central                 |                                             |
| South Atlantic                     |                                             |

*Figure 13: Associations of predictors with violent crime rate*

The intercept of the linear model is 506.422460, meaning that when all predictors in the model are 0, the expected violent crime rate is 506.422460 per 100,000 people. A one SD increase in murder rate is associated with an increase of 125.193698 violent crimes per 100,000 people, holding all other predictors constant. A one SD increase in robbery rate is associated with an increase of 257.019612 violent crimes per 100,000 people, holding all other predictors constant. A one SD increase in population density is associated with an increase of 138.323327 violent crimes per 100,000 people, holding all other predictors constant. States in the Pacific are associated with an increase of 140.403301 violent crimes per 100,000 people compared to states outside of the Pacific, holding all other predictors constant. States in the Mountain region are associated with an increase of 77.552366 violent crimes per 100,000 people, holding all other predictors constant.

Notable observations from this data include the relationships between the gun control laws and violent crime rate as well as HFR and violent crime rate. An increase in the standard deviation of Household Firearm Ratio is associated with an increase in violent crime rate, holding other predictors constant. Additionally, the presence of universal background check laws and shall carry laws are associated with a decrease in violent crime rate, holding other predictors constant. However, the presence of a permit-to-carry law is associated with an increase in violent crime rate. Graphs that model the relationship between each law, HFR, and violent crime rate are attached in the appendix as Figures 5A through 5C. Notably, Simpson's Paradox might be impacting some slopes. The scatterplots show that the presence of a permit to purchase or universal background checks law impacts the correlation (positive or negative) between HFR and violent crime rate. The linear model produced using backwards selection could not capture this relationship. Further exploration using interactions would be important to more fully understand the impact of these laws and HFR on violent crime rate.

One explanation for the relationship between presence of a law and violent crime rate, differing from the effectiveness of the law, may be a pattern among the types of states that adopt each law

(ie. party, demographics, size, etc.). One possible avenue for further exploration of these questions is an investigation into the politically biased interpretation and enforcement of these laws as well as the politically-based social response of state citizens to these laws, and the impact that may have on these patterns. Within the past twenty years, gun control has become increasingly politicized. In this dataset, various gun control laws exist in red and blue states alike, so it would be interesting to see if the politicization of gun control can further reveal or hide patterns about the data using more recent measures.

### ***Robbery Rate Model***

The robbery rate is a measure of the number of robberies occurring in a given year per 100,000 residents of that state. Descriptive statistics summarizing the robbery rate are shown below:

| Minimum | 1st Quartile | Median | Mean  | 3rd Quartile | Maximum |
|---------|--------------|--------|-------|--------------|---------|
| 6.4     | 70.7         | 123.4  | 145.2 | 190          | 684     |

*Figure 14: Descriptive statistics for the robbery rate in the 50 States of the United States from 1980 through 1999 (1980 and 1999 inclusive values).*

As you can see from the large difference of about ~22 more robberies for the mean than for the median, there is a leftward skew for the distribution. The maximum and minimum are very far outside of the  $1.5 \times \text{IQR}$ , so there are definitely outliers for the dataset, and likely many outliers. As I stated earlier, this high variance is reasonable given that these observations are at a state level and the state category is highly non-standard in population, area, and a variety of demographic factors that could relate to the robbery rate.

During the process of inferential exploratory analysis, we arrived at a set of 5 models to examine the relationship between the robbery rate and a variety of potential predictors (5 not 6 models were fitted because insurmountable errors prevented us from running the Gamma Identity Model). Of these models, the most accurate predictive model was the Gaussian Identity Generalized Linear Model ( $\text{MSE} = 24166$ ). There was a slight bow in the distribution of the plot of the residuals, but this relationship didn't appear to be strong enough to invalidate the results of the Linear GLM. The linear model was only slightly better than the other two next best models which each produced MSEs that were only 0.5% and 4.11% higher than our best regression model. Of the predictors used by the linear regression model, our second best model, the gaussian log regression, only weighted murder as one of the top five predictors of robbery rate and the rest of the predictors were different. This large difference in predictors selected depending on link transformation of the Gaussian regression indicated that there are a variety of predictors of robbery rate that have a similarly strong ability to predict this response. Examining the best predictors of the linear regression model, I found violent ( $59.660$ ;  $p < 2e-16$ ), MiddleAtlantic ( $54.006$ ;  $p = 8.32e-15$ ), HFR ( $-41.839$ ;  $p < 2e-16$ ), murder ( $31.648$ ;  $p < 2e-16$ ),

and EastSouthCentral (29.013;  $p = 9.56e-05$ ) were all strongly related to the robbery rate by state. These five predictors each had very significant relationships to the robbery rate ( $p$ -values  $\ll \alpha = 0.0001$ ) and the slopes of these five predictors were the highest in magnitude of any of the linear regression models' predictors. The positive slopes for violent and murder indicated that holding all else constant, there would be an increase in robbery rate of 59.660/100,000 per one sd increase of violent crime rate and an increase in robbery rate of 31.648/100,000 per one sd increase of murder rate. When a state was in the Middle Atlantic or East South Central Region, holding all other predictors constant, there was a much higher robbery rate with the expected value for the robbery rate increasing by 54.006 for Middle Atlantic = True and by 20.013 for East South Central = True. Finally, there was a negative relationship between the HFR and robbery rate indicating that in states one standard deviation higher than average in household firearm ratio, holding all other predictors constant, there would be an expected lower robbery rate by -41.839/100,000.

\*A “-” before the predictor means that the slope was negative:

| Model Type                      | MSE      | Top 5 Predictors High to Low abs(Slope)                                       |
|---------------------------------|----------|-------------------------------------------------------------------------------|
| Linear Regression               | 24166.15 | Violent, MiddleAtlantic, -HFR, murder, EastSouthCentral                       |
| Gaussian Log Regression         | 24228.74 | -NewEngland, -HuntLic, violent, murder, EastNorthCentral                      |
| Gaussian Square Root Regression | 25203.04 | MiddleAtlantic, violent, EastNorthCentral, EastSouthCentral, WestNorthCentral |
| Gamma Square Root Regression    | 26161.52 | Pacific, MiddleAtlantic, violent, Mountain, EastNorthCentral                  |
| Gamma Log Regression            | 36936.37 | Pacific, MiddleAtlantic, EastNorthCentral, WestSouthCentral, Mountain         |

*Figure 15: The MSEs and Top 5 Predictors of the 6 Models Fitted to Predict the Robbery Rate*

| Predictor Variable | Slope   | Standard Error | P-Value for Null |
|--------------------|---------|----------------|------------------|
| Violent            | 59.66   | 2.515          | $< 2e-16$        |
| MiddleAtlantic     | 54.006  | 6.821          | 8.32E-15         |
| HFR                | -41.839 | 3.408          | $< 2e-16$        |
| murder             | 31.648  | 2.741          | $< 2e-16$        |
| EastSouthCentral   | 29.013  | 7.397          | 9.56E-05         |
| HuntLic            | 28.635  | 4.669          | 1.38E-09         |



|          |         |       |          |
|----------|---------|-------|----------|
| Mountain | 26.674  | 5.334 | 7.06E-07 |
| law      | 23.209  | 3.731 | 8.10E-10 |
| male     | -20.514 | 2.538 | 2.44E-15 |
| income   | 20.492  | 2.366 | < 2e-16  |

Figure 16: The ten best predictors and associated slope, standard error, and p-value for the Identity Link Gaussian Distribution Generalized Linear Model predicting the robbery rate based on a backward-selected group of 32 initial predictor variables (all variables are defined in the “Variable Description” section of the appendix).

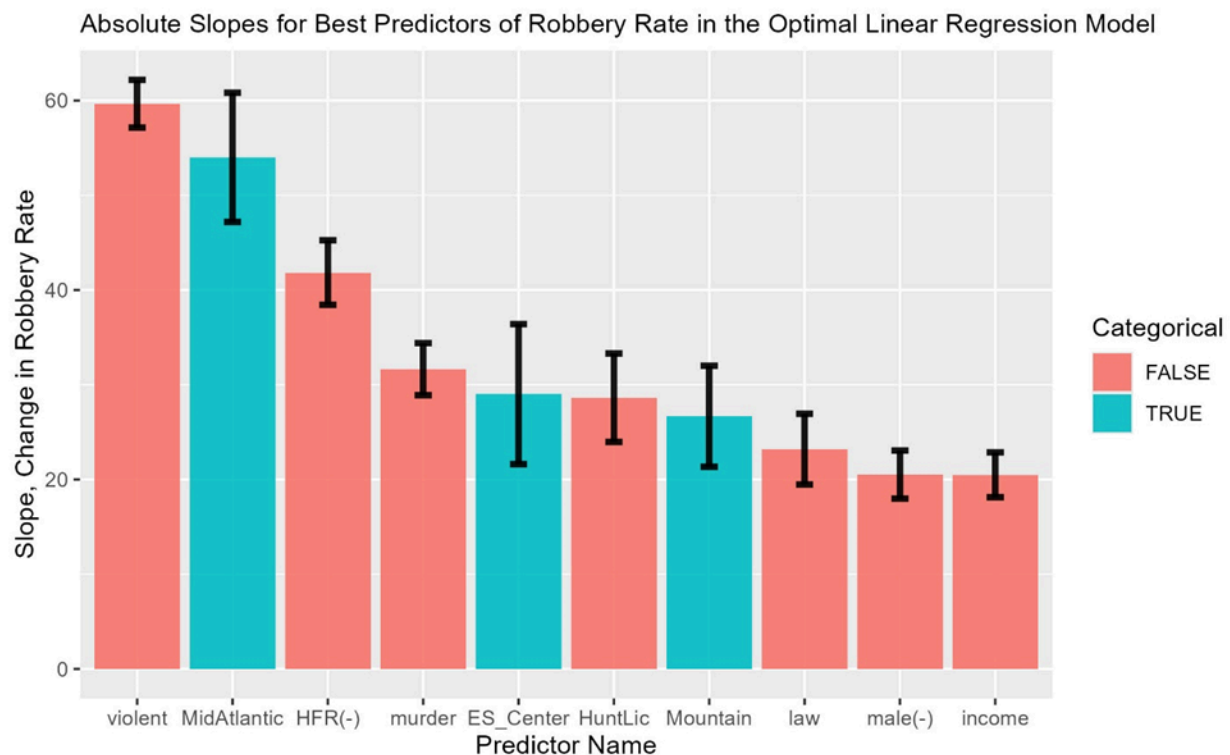


Figure 17: Magnitude of Predictor Slopes for a Gaussian Identity Link Generalized Linear Model Showing Error Bar For SE, Rank of Predictor’s Magnitude, and Colored by whether the Predictor is a Dummy Variable or Numerical Variable

### ***Gun Ownership by State: Household Firearm Ratio (HFR) Model***

The Household Firearm Ratio (HFR) is an estimate of the percentage of households owning guns in each of the states; the numerator is gun owning households and the denominator is the

number of households in the state. As previously mentioned, this variable was generated in a study from RAND based on a set of surveys about household firearm ownership and proxy variables that predict firearm ownership by state. The RAND Study's estimates of HFR were arrived at by creating predictive models and using a method called multilevel regression with poststratification as a way to extrapolate the sparse coverage of existing studies to all of the states in the United States. The response variable for their study was the HFR, which was primarily modeled based on the gun ownership findings of the Behavioral Risk Factor Surveillance System, General Social Survey, and Pew Research Center and Gallup surveys on gun ownership, while the proxy predictors in the RAND survey were a set of demographic data for each of the states (these predictors are listed in the Variables Document in the Appendix). The researchers used this model to impute any missing values of the HFR from 1980 to 2016. Since some of the predictors used to generate HFR were included in the potential predictors of HFR in our analysis and predictive models, these predictor variables are more likely to have an outsized perceived relationship to the response since the response is an estimate approximated as a function of this subset of predictors in our dataset. Descriptive statistics summarizing the HFR are shown below:

| Minimum | 1st Quartile | Median | Mean   | 3rd Quartile | Maximum |
|---------|--------------|--------|--------|--------------|---------|
| 0.075   | 0.423        | 0.5125 | 0.4853 | 0.577        | 0.8     |

*Figure 18: Descriptive statistics for the HFR in the 50 States of the United States from 1980 through 1999 (1980 and 1999 are inclusive years)*

As you can see from the higher absolute difference between median and 1<sup>st</sup> Quartile and the median and 3<sup>rd</sup> Quartile, there is a slight rightward skew to this dataset. The minimum and Maximum values are far outside of the 1.5\*IQR range indicating that this dataset does have some very large outliers. This, however, would be expected since the categorization of “state” has great variance in population, geographic size, and demographic patterns, so its reasonable that HFR would also very greatly from state to state.

Through my inferential exploratory analysis, I arrived at a set of 6 models to examine the relationship between the State Household Firearm Ratio (HFR) and a variety of potential predictors. Of these models, the most accurate predictive model was the Gamma Generalized Linear Model with an identity link transformation. This model was only slightly better than the other two next best models which each produced MSEs that were only 5.45% and 5.5% higher than the Gamma Identity regression model. Examining the best predictors of the Gamma Identity model, I found that “EastSouthCentral” (Beta = 0.08136;  $p < 2e-16$ ), “HuntLic” (Beta = 0.06763;  $p < 2e-16$ ), “SouthAtlantic” (Beta = 0.053092;  $P = 3.25e-16$ ), “murder” (Beta = 0.051148;  $p < 2e-16$ ), and permit (Beta = -0.048100;  $p < 2e-16$ ) were all strongly related to the HFR by state. These five predictors each had very significant relationships to the HFR ( $p$ -values  $\ll \alpha = 0.0001$ ) and the slopes of these five predictors were the highest in magnitude of any of the

models' predictors. The positive slopes for “murder” and “HuntLic” (hunting license) indicated that holding all else constant, for a change of one standard deviation in these predictors, there would be an increase of 0.051 and 0.068 respectively in the HFR. When East South Central and South Atlantic were True (a state was located in these regions as defined in the Regions of the United States Map in the Appendix) holding all else constant, there was an associated 0.081 increase in the HFR (EastSouthCentral) and a 0.053 increase in the HFR (SouthAtlantic). Finally, if the state had a gun permitting requirement, holding all else constant, there would be a lower HFR for the state by 0.048100.

\*A “-” before the predictor means that the slope was negative

| Model Type                      | MSE        | Top 5 Predictors High to Low abs(Slope)                     |
|---------------------------------|------------|-------------------------------------------------------------|
| Gamma Identity Regression       | 0.04409864 | EastSouthCentral, HuntLic, SouthAtlantic, murder, & -permit |
| Gaussian Log Regression         | 0.04650305 | Male_FS_S, -permit, -density, HuntLic, EastNorthCentral     |
| Linear Regression               | 0.0465654  | Male_FS_S, HuntLic, GunsAmmo, -permit, EastSouthCentral     |
| Gaussian Square Root Regression | 0.04678685 | Male_FS_S, HuntLic, -permit, -density, EastSouthCentral     |
| Gamma Square Root Regression    | 0.0475362  | Male_FS_S, HuntLic, -permit, GunsAmmo, -density             |
| Gamma Log Regression            | 0.04817524 | Male_FS_S, -permit, HuntLic, -density, EastNorthCentral     |

*Figure 19: The MSEs and Top 5 Predictors of the 6 Models Fitted to Predict the HFR*

| Predictor Variable | Slope     | Standard Error | P-Value for Null |
|--------------------|-----------|----------------|------------------|
| EastSouthCentral   | 0.08136   | 0.009209       | < 2e-16          |
| HuntLic            | 0.06763   | 0.00573        | < 2e-16          |
| SouthAtlantic      | 0.053092  | 0.006362       | 3.25E-16         |
| murder             | 0.051148  | 0.003365       | < 2e-16          |
| permit             | -0.0481   | 0.0055         | < 2e-16          |
| GunsAmmo           | 0.042068  | 0.002891       | < 2e-16          |
| MiddleAtlantic     | 0.037201  | 0.006907       | 9.56E-08         |
| WestSouthCentral   | 0.036317  | 0.00921        | 8.76E-05         |
| robbery            | -0.034387 | 0.003564       | < 2e-16          |

|         |           |          |         |
|---------|-----------|----------|---------|
| density | -0.024314 | 0.002176 | < 2e-16 |
|---------|-----------|----------|---------|

Figure 20: The ten best predictors and associated slope, error, deviance, and p-value for the Identity Link Gamma Distribution Generalized Linear Model predicting Household Firearm Ratio (HFR) Based on a backward-selected group of 32 initial predictor variables (all variables are defined in the “Variable Description” section of the appendix).

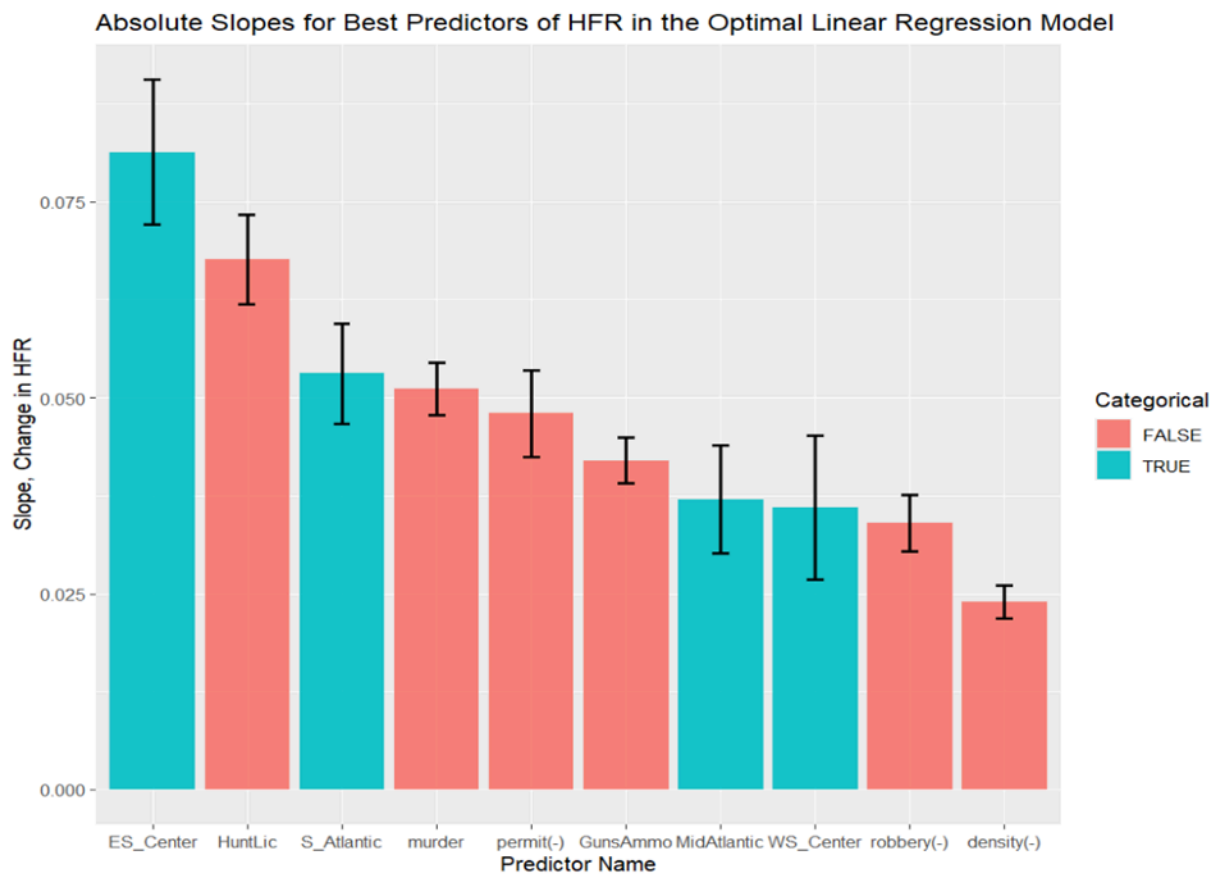


Figure 21: Magnitude of Predictor Slopes for a Gamma Identity Link Generalized Linear Model Showing Error Bar for SE, Rank of Predictor’s Magnitude, and Colored by whether the Predictor is a Dummy Variable or Numerical Variable

## Conclusion

### Limitations

One major limitation of our modeling for interpretation was the exclusion of interactions between variables. Inclusion of interactions would have helped increase the flexibility of the models; flexibility would be a useful adaptation to the dataset given the high variance caused by

the multicollinearity between several of the variables (afam and cauc; HFR and Male\_FS\_S). The MSE for the KNN models shows that more flexible fits would be appropriate for this data. Future studies should explore interactions between predictors when fitting and interpreting models in order to better understand the influence of each predictor. We did not include predictor variables that were sensitive to the level of urbanization in a given state. A lot of the relationships we found might be explained by an urbanization variable that hasn't been included in our dataset.

### ***Discussion***

A main takeaway from the classification models is that a state's region is highly influential in whether or not any of the investigated gun laws had been passed in that state. Additionally, we found that lasso logistic regression produced the highest AUC for two out of the three gun laws we modeled the presence of. Thus, if our goal were purely to predict whether or not a state had a gun law present, lasso could be a good predictive method. However, to use this statement beyond conjecture purposes, we would need to increase the sample size and model the presence/absence of a greater number of gun regulation laws against our predictors. Takeaways from regression models are that crime rates are highly correlated with each other, but that crime rates generally cannot be predicted strongly by gun legislation or Household Firearms Ratio (HFR) (with the one exception of murder to HFR association; Fig. 20). Gun ownership (Fig. 21) and crime rates (Fig. 10, Fig. 13, Fig. 16) are predicted well by region. HFR was moderately, negatively predicted by the presence of a permit law in states which could be related to distinct trends in each variable depending on rural or urban region, but more accurate predictors or rural vs. urban population would be needed to see if this is the case. To draw further conclusions from regression models about the effectiveness of gun legislation or the influence of gun ownership on crime would require more extensive studies and modeling that more carefully examine the influence of other characteristics of states on legislation.

### ***Areas for Future Research***

In addition to the ideas for future methods we have already touched on, for both classification and regression models, one area for further research includes creating better methods to limit variance inflation (VIF) values. In this project, we mitigated high variance inflation by removing variables that had high initial VIFs, given that they were not too significant to the model or interesting to the problem. This turned out to be quite a subjective method. Further steps could include Principal Component Analysis to combine predictors that we found to be highly collinear so that we can both include interesting and significant predictors as well as limit variance inflation. Additionally, future research could further explore the rural/urban divide by using a more specific measure for population density and explore regional differences using a by group analysis.

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[https://www2.census.gov/geo/pdfs/maps-data/maps/reference/us\\_regdiv.pdf](https://www2.census.gov/geo/pdfs/maps-data/maps/reference/us_regdiv.pdf)

## Appendix

Figure 1A. Variable Names and Descriptions

| year             | Calendar year in which the set of observations was collected                                                                                            |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| violent          | Rate of violent crime (assaults, etc.) cases of violence that did not lead to the death of the victim, per 100,000 people                               |
| murder           | Rate of violence or negligence that did lead to death (murder, manslaughter, etc.), per 100,000 people                                                  |
| robbery          | Rate of robberies per 100,000 people                                                                                                                    |
| prisoners        | Incarceration rate in the state in the previous year (sentenced prisoners per 100,000 residents; value for the previous year).                          |
| afam             | Percent of state population that is African-American, ages 10 to 64.                                                                                    |
| cauc             | Percent of state population that is Caucasian, ages 10 to 64.                                                                                           |
| male             | Percent of state population that is male, ages 10 to 29                                                                                                 |
| population       | State population, in millions of people.                                                                                                                |
| income           | Real per capita personal income in the state (US dollars). (mean)                                                                                       |
| density          | Population per square mile of land area, divided by 1,000                                                                                               |
| law              | Is there a shall carry law in effect that year in the state?                                                                                            |
| HFR              | Household Firearm Ratio                                                                                                                                 |
| universl         | State has a universal background checks law.                                                                                                            |
| permit           | Presence of permit to purchase law (1=yes). Is there any law in the state that requires people to get a permit for a handgun before purchasing the gun. |
| Female_FS_S      | Female Firearm Suicide/Total Female Suicide *100                                                                                                        |
| Male_FS_S        | Male Firearm Suicide/Total Male Suicide *100                                                                                                            |
| Hunt_Lic         | Square root of Hunting Licenses/Population. Rate per capita.                                                                                            |
| GunsAmmo         | Within-year standardization of the square root of Guns and Ammo (magazine) Subscriptions/Population.                                                    |
| Pacific          | Alaska, California, Hawaii, Oregon, Washington                                                                                                          |
| Mountain         | Arizona, Colorado, Idaho, New Mexico, Montana, Utah, Nevada, Wyoming                                                                                    |
| WestNorthCentral | Iowa, Kansas, Minnesota, Missouri, Nebraska, North Dakota, South Dakota                                                                                 |
| EastNorthCentral | Indiana, Illinois, Michigan, Ohio, Wisconsin                                                                                                            |
| WestSouthCentral | Arkansas, Louisiana, Oklahoma, Texas                                                                                                                    |
| EastSouthCentral | Alabama, Kentucky, Mississippi, Tennessee                                                                                                               |
| South Atlantic   | Delaware, Florida, Georgia, Maryland, North Carolina, South Carolina, Virginia, West Virginia                                                           |
| MiddleAtlantic   | New Jersey, New York, Pennsylvania                                                                                                                      |

|            |                                                                                  |
|------------|----------------------------------------------------------------------------------|
| NewEngland | Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, Vermont          |
| West       | States is in the Pacific or Mountain region                                      |
| South      | State is in the South Atlantic, East South Central, or West South Central Region |
| Midwest    | State is in the East North Central or West North Central Region                  |
| Northeast  | State is in New England or Middle Atlantic                                       |
|            |                                                                                  |

Figure 2A. Summary Values for Linear Regression predicting Murder Rate

| term                | estimate   | std.error  | statistic | p.value  |
|---------------------|------------|------------|-----------|----------|
| (Intercept)         | 7.8179928  | 0.1128224  | 69.294686 | 0.00E+00 |
| violent             | 1.7705967  | 0.16079634 | 11.011424 | 2.42E-26 |
| robbery             | 2.3448012  | 0.18501648 | 12.673472 | 1.18E-33 |
| prisoners           | -0.2532408 | 0.119815   | -2.113599 | 3.49E-02 |
| afam                | 1.1968324  | 0.08373566 | 14.292983 | 2.08E-41 |
| male                | 0.5220404  | 0.07540681 | 6.922987  | 9.13E-12 |
| density             | 1.0408354  | 0.51664439 | 2.014607  | 4.43E-02 |
| Male_FS_S           | 1.8681334  | 0.08633221 | 21.638891 | 5.51E-82 |
| universl_X1         | 0.4117954  | 0.16040462 | 2.567229  | 1.04E-02 |
| Pacific_X1          | -1.3232955 | 0.21581933 | -6.131497 | 1.37E-09 |
| Mountain_X1         | -0.4618726 | 0.1745232  | -2.646482 | 8.29E-03 |
| WestNorthCentral_X1 | -0.7702848 | 0.17930696 | -4.2959   | 1.96E-05 |
| WestSouthCentral_X1 | 1.043652   | 0.21591782 | 4.833561  | 1.61E-06 |
| SouthAtlantic_X1    | -1.3680404 | 0.17737258 | -7.712807 | 3.69E-14 |
| MiddleAtlantic_X1   | -0.8976962 | 0.25879968 | -3.468691 | 5.51E-04 |

Figure 3A. Presence of Universal Background Checks Law vs Murder Rate



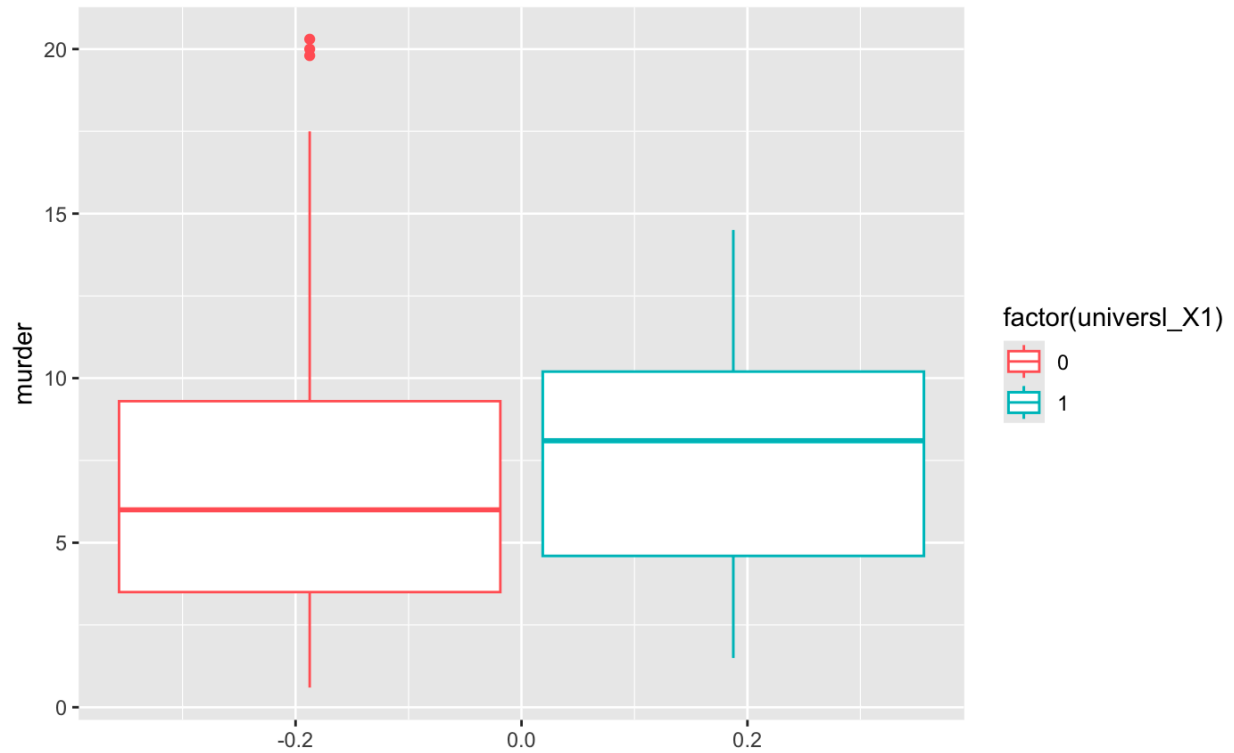


Figure 4A. Summary Values for Linear Regression predicting Violent Crime Rate

| term        | estimate   | std.error | statistic | p.value   |
|-------------|------------|-----------|-----------|-----------|
| (Intercept) | 506.42246  | 8.979745  | 56.396087 | 1.12E-278 |
| year        | 51.978326  | 8.081222  | 6.431989  | 2.18E-10  |
| murder      | 125.193698 | 15.327237 | 8.168054  | 1.24E-15  |
| robbery     | 257.019612 | 11.556179 | 22.240881 | 2.08E-85  |
| prisoners   | 63.012409  | 8.774999  | 7.180902  | 1.61E-12  |
| afam        | -41.249637 | 6.181773  | -6.672785 | 4.73E-11  |
| income      | -26.614686 | 7.185289  | -3.704052 | 2.27E-04  |
| density     | 138.323327 | 38.26163  | 3.615197  | 3.19E-04  |
| HFR         | 38.611369  | 11.24571  | 3.433431  | 6.27E-04  |
| universl1   | -42.972804 | 12.613434 | -3.406908 | 6.90E-04  |
| permit1     | 57.361183  | 14.408756 | 3.980995  | 7.50E-05  |
| Fem_FS_S    | -7.585435  | 3.923362  | -1.933402 | 5.35E-02  |
| HuntLic     | -68.797865 | 12.426643 | -5.536319 | 4.21E-08  |
| GunsAmmo    | -20.088383 | 7.000464  | -2.869579 | 4.22E-03  |

|                   |            |           |           |          |
|-------------------|------------|-----------|-----------|----------|
| Pacific1          | 140.403301 | 15.864466 | 8.850175  | 5.70E-18 |
| Mountain1         | 77.552366  | 11.815456 | 6.563637  | 9.52E-11 |
| WestNorthCentral1 | 33.645114  | 12.32615  | 2.729572  | 6.48E-03 |
| EastSouthCentral1 | -49.028908 | 15.426384 | -3.17825  | 1.54E-03 |
| SouthAtlantic1    | 55.458528  | 11.874236 | 4.670492  | 3.53E-06 |
| MiddleAtlantic1   | -72.086468 | 17.175834 | -4.196971 | 3.01E-05 |
| law_yes           | -44.737585 | 9.463063  | -4.727601 | 2.69E-06 |

Figure 5A. Household Firearm Ratio vs Violent Crime Rate colored by Presence of Universal Background Checks Law

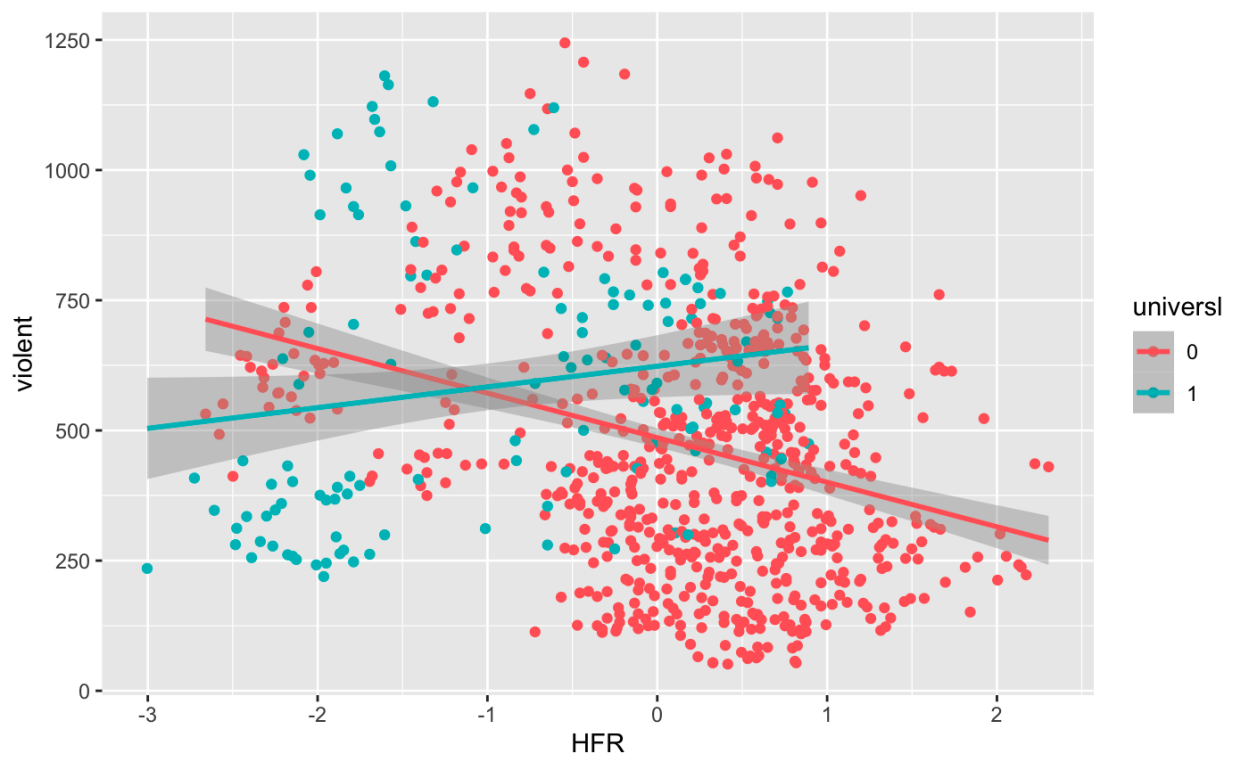


Figure 5B. Household Firearm Ratio vs Murder Rate colored by Presence of Permit to Purchase Law



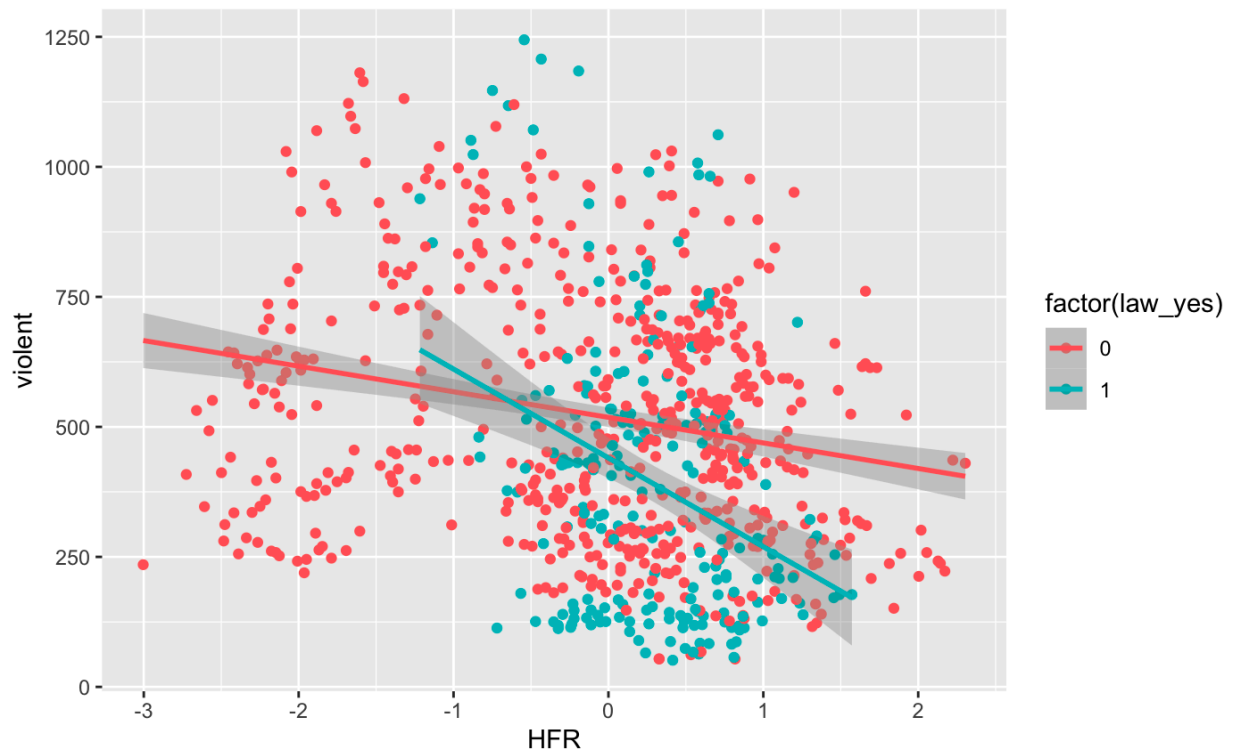


Figure 6A: Predictors positively and negatively associated with universal background check presence as determined by logistic regression models

| Predictors associated with an increased chance of a state having universal background check laws | Predictors associated with a decreased chance of a state having universal background check laws |
|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| murder                                                                                           | year                                                                                            |
| afam                                                                                             | violent                                                                                         |
| population                                                                                       | male                                                                                            |
| HuntLic                                                                                          | income                                                                                          |
| GunsAmmo                                                                                         | HFR                                                                                             |
| permit                                                                                           | Fem_FS_S                                                                                        |
| East SouthCentral                                                                                | shall_carry                                                                                     |
|                                                                                                  | mountain                                                                                        |

|  |                  |
|--|------------------|
|  | EastNorthCentral |
|  | MiddleAtlantic   |

Table 7A: Predictors positively and negatively associated with shall carry law presence as determined by logistic regression models

| Predictors associated with an increased chance of a state having shall-carry laws | Predictors associated with a decreased chance of a state having shall-carry laws |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| year                                                                              | violent                                                                          |
| robbery                                                                           | murder                                                                           |
| male                                                                              | afam                                                                             |
| pacific                                                                           | population                                                                       |
| MiddleAtlantic                                                                    | income                                                                           |
| NewEngland                                                                        | density                                                                          |
|                                                                                   | Fem_FS_S                                                                         |
|                                                                                   | permit                                                                           |
|                                                                                   | Mountain                                                                         |
|                                                                                   | WestSouthCentral                                                                 |
|                                                                                   | EastSouthCentral                                                                 |
|                                                                                   | WestNorthCentral                                                                 |

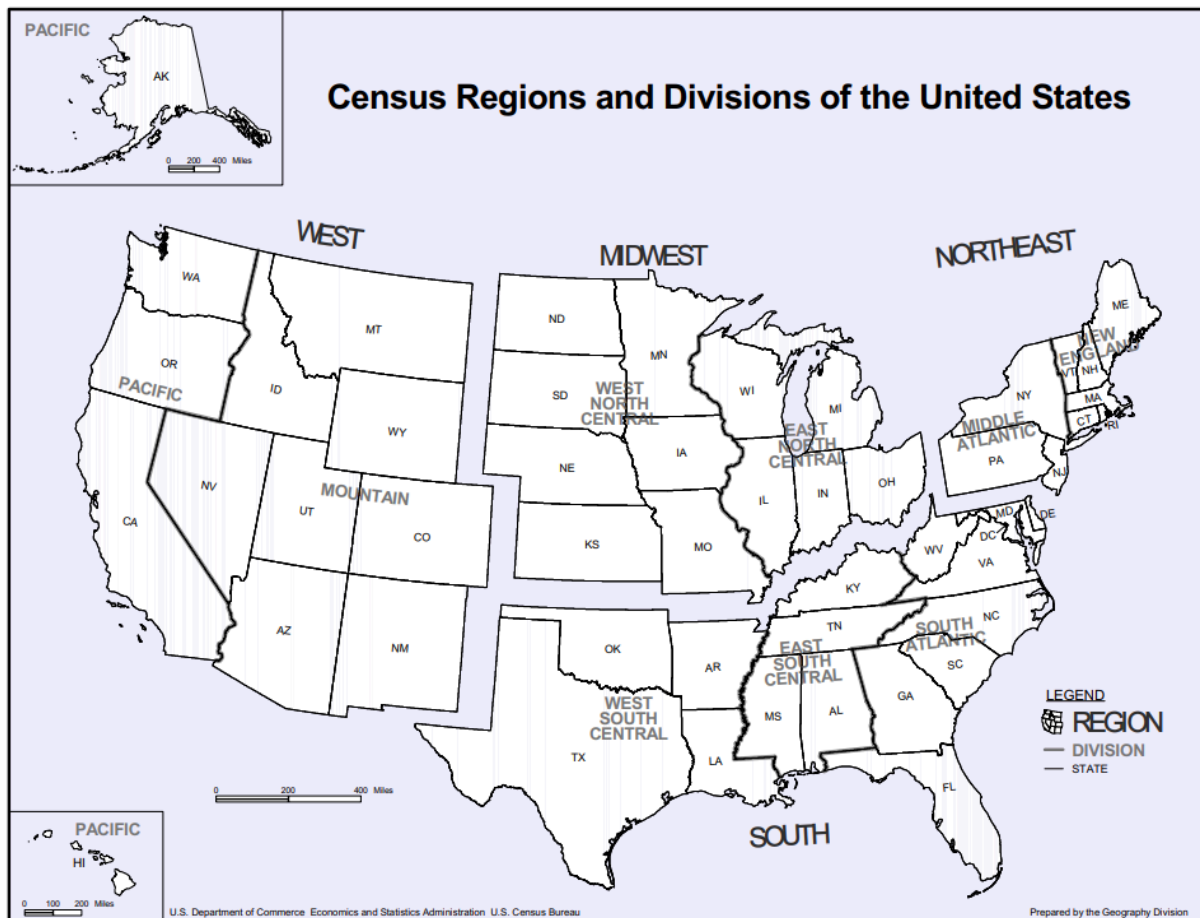
Table 8A: Predictors positively and negatively associated with shall carry law presence as determined by logistic regression models

| Predictors associated with an increased chance of a state having permit-to-purchase laws | Predictors associated with a decreased chance of a state having permit-to-purchase laws |
|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| year                                                                                     | income                                                                                  |
| violent                                                                                  | HFR                                                                                     |

|                  |                  |
|------------------|------------------|
| afam             | Pacific          |
| Fem_FS_S         | Mountain         |
| universl         | WestSouthCentral |
| WestNorthCentral | EastSouthCentral |
|                  | SouthAtlantic    |
|                  | MiddleAtlantic   |

[Link to Census Regions and Divisions of the United States Map](https://cctigers-my.sharepoint.com/:b:/g/personal/l_hartman2023_coloradocollege_edu/ETcppS1MWItJlwZdxNiToiAB1VOuvOqSfVsKBWV2nbD28A?e=OasAGa)

[https://cctigers-my.sharepoint.com/:b:/g/personal/l\\_hartman2023\\_coloradocollege\\_edu/ETcppS1MWItJlwZdxNiToiAB1VOuvOqSfVsKBWV2nbD28A?e=OasAGa](https://cctigers-my.sharepoint.com/:b:/g/personal/l_hartman2023_coloradocollege_edu/ETcppS1MWItJlwZdxNiToiAB1VOuvOqSfVsKBWV2nbD28A?e=OasAGa)



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[Link to html file for classification tasks code](#)

[https://drive.google.com/file/d/1ZNENAzOyGpEYEIB\\_5\\_YY\\_2paLJbGhZ0l/view?usp=sharing](https://drive.google.com/file/d/1ZNENAzOyGpEYEIB_5_YY_2paLJbGhZ0l/view?usp=sharing)

[Link to html file for Regression - Response is Murder Rate](#)

<https://drive.google.com/file/d/1oHdX-iQ36L4wUIeozigMp80dSEYvtivn/view?usp=sharing>

[Link to html file for Regression - Response is Violent Crime Rate](#)

<https://drive.google.com/file/d/1Q8z3J6-5kOk58dTm2sEWpVnlPTOWkbR8/view?usp=sharing>

[Link to html file for Regression - Response is Robbery Rate](#)

[https://cctigers-my.sharepoint.com/:u:/g/personal/l\\_hartman2023\\_coloradocollege\\_edu/ESQrcrAsVwFFozH3qd6NmQwBXfAJhxAfBEPNO\\_pI2lxYWA?e=vcG4nj](https://cctigers-my.sharepoint.com/:u:/g/personal/l_hartman2023_coloradocollege_edu/ESQrcrAsVwFFozH3qd6NmQwBXfAJhxAfBEPNO_pI2lxYWA?e=vcG4nj)

[Link to html file for Regression - Response is Household Firearm Ratio](#)

[https://cctigers-my.sharepoint.com/:u:/g/personal/l\\_hartman2023\\_coloradocollege\\_edu/Efn5VS7adYxKqdjSI9x\\_HYBNPu6V1\\_I2qFZSXI3ifDrhw?e=PYtFFs](https://cctigers-my.sharepoint.com/:u:/g/personal/l_hartman2023_coloradocollege_edu/Efn5VS7adYxKqdjSI9x_HYBNPu6V1_I2qFZSXI3ifDrhw?e=PYtFFs)